Courtney Green

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EDUCATION

Georgetown University, Graduate School of Arts & Sciences, Washington, D.C. May 2026
Master of Science in Data Science and Analytics

- **GPA:** 4.0/4.0
- **Relevant Coursework:** Probabilistic Modeling & Statistical Computing, Database Systems & SQL, Neural Nets & Advanced Deep Learning, Causal Inference for Computational Social Science, Reinforcement Learning, Big Data & Cloud Computing, Machine Learning for App Development

University of Virginia, Frank Batten School of Leadership and Public Policy, Charlottesville, VA May 2024
Bachelor of Arts in Public Policy and Leadership; Minor in Computer Science

TECHNICAL SKILLS

- **Languages & Libraries**: Python (NumPy · Pandas · Scikit-learn · PyTorch · TensorFlow · Keras · Matplotlib · Seaborn · FAISS · FastAPI · Streamlit · BeautifulSoup) · R · SQL · Java · JavaScript · React · HTML/CSS
- **Infrastructure & Tools:** AWS · Azure · Docker · PostgreSQL · GitHub · Tableau · Power BI · Stata · Excel · JupyterLab
- **Methods:** Machine Learning · Causal Inference · Statistical Modeling · Data Wrangling · ETL · Database Management · Data Visualization · Feature Engineering · Natural Language Processing · Reinforcement Learning

RELEVANT EXPERIENCE

The Associated Press, Remote October 2025 – Present
Elections Web Scraper

- Collect and verify live election results for AP across 20+ counties using Python, supporting 6+ elections across 2025–2026.
- Monitor live scrapes in real time on election nights to ensure data accuracy and flag discrepancies, contributing to AP's trusted election coverage.

Georgetown Prisons and Justice Initiative (PJI), Washington, D.C. October 2024 – May 2026
Scholars Program Associate

- Maintained and organized complex data sets related to program operations and performance; conducted data analysis to inform program development and reported key performance metrics to the DC Department of Corrections (DOC).
- Assisted in creating and refining database structures and processes for more effective data collection and visualization, contributing to long-term program sustainability and data-driven decision-making.

Georgetown Data Science & Analytics, Washington, D.C. May 2025 – December 2025
Graduate Teaching Assistant

- Causal Inference for Computational Social Science – Supported instruction on methods including propensity score matching, synthetic controls, instrumental variables, causal forests, and double-debiased machine learning. Collaborated with the professor to develop labs, materials, and review resources for final projects.
- Probabilistic Modeling and Statistical Computing – Assisted with a graduate-level course covering probability theory, Monte Carlo simulation, Bayesian models, EM, hidden Markov models, and graphical models; collaborated with the professor to develop course materials, held office hours, and supported students in R.

Guidehouse, Arlington, VA June 2025 – August 2025
Commercial Sustainability Intern

- Developed and refined an internal scoring tool to evaluate corporate CDP (Carbon Disclosure Project) responses; implemented complex formulas and conditional logic to automate scoring, flag data gaps, and generate dashboards for climate disclosure performance assessment.
- Led the design of an AI-powered Retrofit Portfolio Ranking Engine prototype, applying predictive modeling to help affordable housing agencies prioritize retrofit projects based on emissions impact, deferred maintenance needs, and funding alignment.
- Supported a 20-year energy demand forecast for NYISO using time series and ensemble models (ARIMA, linear regression, random forest) in R; contributed to model tuning and analysis of demand, wind, and nuclear generation data.
- Analyzed downscaled climate models (e.g., CMIP6 variants) to assess fitness for localized climate risk forecasting; compared model resolution, variable treatment, and performance across key climate indicators.

EVgo, Los Angeles, CA June 2024 – August 2024
Market Development & Public Policy Intern

- Researched and analyzed government incentives, including the 30C tax credit and NEVI program funds, to support the development of electric vehicle charging infrastructure.
- Drafted and optimized data-driven reports, evaluating electricity rate designs and charging infrastructure development.
- Collaborated on cross-functional teams, working closely with software developers, policy experts, and data analysis teams to support infrastructure expansion through the Department of Energy's Title 17 Clean Financing program.

VOX Global, Washington, D.C. May 2023 – August 2023
Public Affairs Intern

- Performed comprehensive audits and data collection to inform client reports, using analytical tools like Excel and Tableau to identify trends in corporate social responsibility (CSR) initiatives.
- Developed media outreach strategies with team members, managing stakeholder engagement data, and supporting decision-making for Fortune 500 clients.
- Wrote, edited, and proofread various reports and internal communications, helping streamline client information-sharing processes.

Mindset, Washington, D.C. June 2021 – August 2022
Policy Research Intern
- Performed qualitative and quantitative research on cybersecurity and privacy, analyzing policy frameworks and conducting risk assessments to provide actionable recommendations for clients.
- Tracked legislative trends in cybersecurity and privacy to assess the potential impacts of regulatory changes for clients.
- Created and regularly maintained an extensive cybersecurity and privacy legislative database of congressional floor activity, the Biden Administration, and independent regulators for clients.

PRESENTATIONS & AWARDS

DC Metro StatConnect Graduate Research Poster Competition (2025)
George Mason University
- Awarded 1st place in the MS category for outstanding research contributions and poster presentation; recognized for *Data vs. Dogma: Examining the Intersection of Abortion Policy and Maternal Health Disparities in the U.S.*

Data Science & Analytics Returning Student Scholarship (2025)
Georgetown University
- Awarded 2nd place in a competitive merit-based scholarship granted to 38% of continuing students through anonymous faculty review of an independent data visualization project; recognized for *Humanitarian Shield: A Data-Driven Exploration of Humanitarian Aid Worker Vulnerability*.

University Achievement Award Scholarship (2020 – 2024)
University of Virginia
- Merit-based scholarship recognizing academic excellence, leadership, and citizenship; awarded to 40–50 students per graduating class.

Future Scholarship (2024)
Palantir
- Merit-based scholarship supporting students from underrepresented racial and ethnic communities in STEM.

PROJECTS

Quantitative Analysis of Maternal Health Indicators and Reproductive Policy Variation
Technology Stack: R (tidyverse, ggplot2, stats, logistf), Quarto, Git
- Compiled and cleaned multi-source datasets (CDC, BLS, KFF, Guttmacher, peer-reviewed journals); standardized formats, recoded categorical variables, and merged 100+ indicators across 52 jurisdictions.
- Applied non-parametric and exact statistical tests (Permutation ANOVA, Kruskal–Wallis, Fisher's Exact, Exact Logistic Regression) to evaluate associations between maternal health metrics and post-Dobbs abortion policy levels.
- Visualized ranked, categorical, and binary outcome distributions to identify and compare significant differences across uneven policy-level sample sizes.

Probabilistic Modeling of Prosecutorial Misconduct and Over-Policing in Chicago
Technology Stack: Python (Pandas, Scikit-learn, GeoPandas, Plotly), Quarto, Git, PCA, t-SNE, DBSCAN
- Cleaned and merged exoneration, arrest, incarceration, and misconduct datasets; engineered a counterfactually balanced dataset for supervised modeling.
- Trained logistic regression and random forest classifiers to predict wrongful conviction risk (ROC-AUC: 0.95); applied PCA and t-SNE for dimensionality reduction.
- Used K-Means and DBSCAN to identify hidden spatial and demographic patterns; built a reproducible Quarto site to visualize geospatial and clustering results.

Modeling Voter Turnout and Partisan Lean in the 2024 U.S. Presidential Election
Technology Stack: Python (Scikit-learn, Pandas, Matplotlib), R, Git, Logistic Regression, SVM, KNN, Random Forest
- Cleaned and merged county-level demographic and election data; engineered features to classify partisan lean (Democrat-leaning vs. non-Democrat-leaning) and voter turnout (high vs. low).
- Trained and compared classification models (Logistic Regression, SVM, KNN, Random Forest) using best subset selection, cross-validation, and validation curves.
- Simulated election-night reporting dynamics using early vs. late-reporting states; evaluated model performance using accuracy, precision, recall, and ROC-AUC.

Predicting Religious Affiliation from Structured Survey Data Using Neural Networks
Technology Stack: Python (TensorFlow, Keras, Scikit-learn), NLTK, Matplotlib, Git
- Preprocessed 21,764 structured survey responses via label encoding, feature scaling, and selection of the top 50 predictors for modeling.

- Trained deep feedforward neural networks with dropout regularization for multi-class classification; benchmarked against logistic regression, SVM, and random forest.
- Evaluated models using cross-validation, confusion matrices, and F1 scores; visualized training dynamics to guide tuning.

Geospatial and Demographic Analysis of U.S. Overdose Crisis
Technology Stack: Quarto, VegaLite, Plotly, Altair, Folium, Python (Pandas), R
- Built an interactive dashboard to analyze county- and state-level overdose mortality using CDC, Census, and USDA data.
- Designed visualizations to highlight geographic clustering, socioeconomic disparities, and demographic patterns in overdose deaths.
- Developed interactive filters enabling exploration of relationships between overdose rates and factors such as income, education, race, and age.
- Applied exploratory data analysis and visual storytelling techniques to support public health and policy insights.

Humanitarian Shield: A Data-Driven Exploration of Humanitarian Aid Worker Vulnerability
Technology Stack: Python (Pandas, Scikit-learn, GeoPandas, Plotly, Matplotlib, Seaborn), Quarto, Git
- Analyzed geographic and temporal patterns of violence against humanitarian aid workers using the Aid Worker Security Database (AWSD), examining victim profiles, attack types, and fatality predictors across crisis regions.
- Applied exploratory analysis, geospatial mapping, and statistical tests to identify critical vulnerabilities and disparities between national and international staff security outcomes.
- Developed an interactive dashboard visualizing spatial, temporal, and demographic patterns of security incidents to support improved security risk management practices for humanitarian organizations.

Style Synth: ML-Powered Fashion Recommendation System
Technology Stack: Python (ResNet-50, FAISS, FastAPI, Streamlit, PostgreSQL, Docker), AWS S3, Prometheus, Grafana
- Built an end-to-end fashion recommendation system using ResNet-50 to extract 2048-dimensional visual embeddings from clothing images, enabling similarity-based outfit generation across occasions and seasons.
- Applied PCA for dimensionality reduction (2048→128D), achieving 16x compression while retaining ~95% of variance; benchmarked FAISS index configurations, reaching sub-millisecond query latency at 517 q/s.
- Designed a full-stack architecture spanning a Streamlit frontend, FastAPI backend, PostgreSQL vector storage, and Docker Compose orchestration across six containerized services.
- Implemented Prometheus and Grafana monitoring to track ML inference latency, API performance, and AWS S3 operation times in production.

Understanding Online Moral Judgment at Scale: NLP & ML Analysis of r/AmItheAsshole
Technology Stack: Python (PySpark, NLTK, Scikit-learn, TF-IDF), AWS (EC2, S3), Apache Spark
- Processed and analyzed 34.5M+ Reddit posts using distributed computing on an Apache Spark cluster hosted on AWS EC2, applying EDA, NLP, and ML across a 14-month dataset.
- Built an NLP pipeline including tokenization, stop word removal, sentiment analysis, and LDA topic modeling to identify linguistic and thematic patterns across moral judgment categories.
- Trained logistic regression and random forest classifiers to predict YTA vs. NTA verdicts; achieved 79.5% F1 score and 83.2% ROC-AUC using post length, sentiment, and TF-IDF features.
- Identified key lexical predictors of verdicts and demographic patterns in moral judgment, finding that longer, more detailed posts with positive sentiment consistently favored NTA outcomes.